

# Pandas: DataFrames + Series

There are two important object types in the pandas module that we will be using throughout the semester to hold tabular datasets. A **DataFrame** is used to contain an entire tabular dataset, often created with `pd.read_csv`. A Series object is used to store the individual columns of a DataFrame.

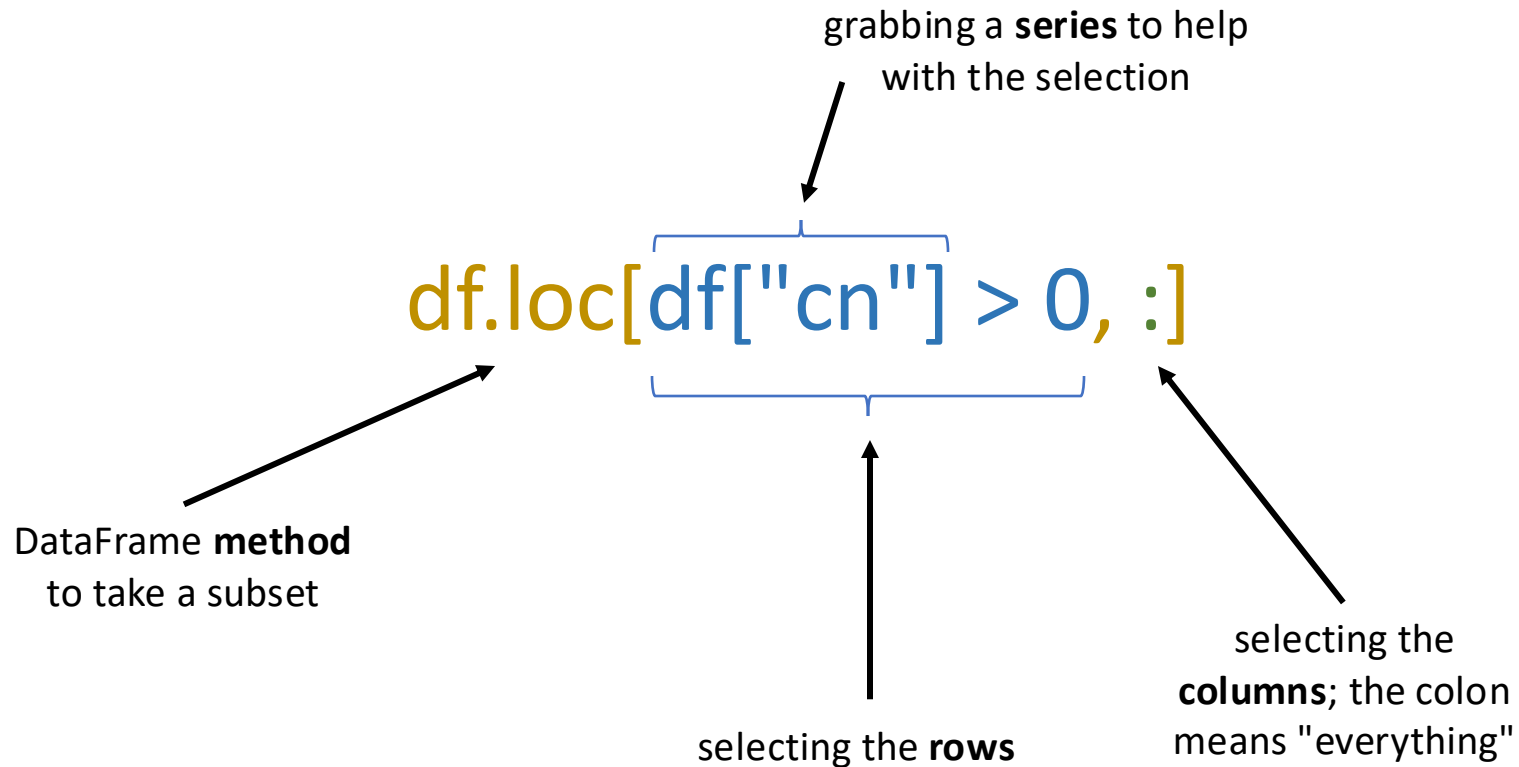
We can access a series through the following syntax, where we assume we a dataset named **df** and a column named **cn**:

```
df["cn"]
```

Grabbing individual series is useful for a number of applications, such as counting the values within the column and using the series as a way of selecting rows in the larger DataFrame object.

# Selecting Rows + Columns

Assuming that we have a DataFrame object, we can select a subset of the rows and columns using the .loc method followed by a special syntax with square brackets.



# Back to Graphics

Now that we have ways of making new datasets, let's go back to our method of creating graphics to see something new. Before, we always used the same dataset for every layer in our plots. But, we can override the default dataset (the one in the `ggplot` function) by adding a *data* argument to any geometry. For example:

```
(ggplot(df) +  
  geom_point(aes(x = "calories", y = "total_fat"), color="red") +  
  geom_point(aes(x = "calories", y = "total_fat"), color="blue", data=df_new))
```

This will create one plot with two sets of points, one in red and one in blue, from two different datasets. Where have we seen plots like this already?